

Let $L \subseteq \Sigma^*$. Define $C_L: \Sigma^* \rightarrow \mathcal{P}(\Sigma^* \times \Sigma^*)$ by

$$C_L(y) = \{ (x, z) \mid xyz \in L \}.$$

Let $\sim \subseteq \Sigma^* \times \Sigma^*$ be an equivalence relation on Σ^* . We say that $L \subseteq \Sigma^*$ is $\sim$ -substitutable if the following condition holds:

$$\forall y_1 \in \Sigma^* \forall y_2 \in \Sigma^* (y_1 \sim y_2 \rightarrow (C_L(y_1) \cap C_L(y_2) = \emptyset \vee C_L(y_1) = C_L(y_2))).$$

Exercise 1. Give an equivalence relation $\sim$ such that $\sim$ -substitutability coincides with k, l -substitutability.

Exercise 2. Modify Yoshinaka's learning algorithm k, l -SGL to obtain a learning algorithm for $\sim$ -substitutable languages. Assuming that the question " $y_1 \sim y_2$?" is decidable in polynomial time, show that the algorithm updates its hypothesis in polynomial time.

We say that $\sim$ is of *finite index* if the number of equivalence classes of $\sim$ is finite.

An *extended context-free grammar* $G = (N, \Sigma, P, \mathcal{S})$ is like a context-free grammar except that instead of a single start nonterminal, it has a *set* of start nonterminals. The language of G is $L(G) = \{ w \in \Sigma^* \mid S \Rightarrow_G^* w \text{ for some } S \in \mathcal{S} \}$.

Problem 3. Let $\sim \subseteq \Sigma^* \times \Sigma^*$ be an equivalence relation of finite index such that each of its equivalence classes is regular. Show that for any context-free grammar $G = (N, \Sigma, P, S)$, there is an equivalent extended context-free grammar $G' = (N', \Sigma, P', \mathcal{S}')$ such that for every $A \in N'$,

$$A \Rightarrow_{G'}^* y_1 \wedge A \Rightarrow_{G'}^* y_2 \quad \text{implies} \quad y_1 \sim y_2 \quad (\dagger)$$

for all $y_1, y_2 \in \Sigma^*$.

(*Hint:* For each equivalence class of $\sim$, take a finite monoid recognizing it. Use the product monoid of these monoids in Problem 5 of Lecture 1, *Formal Grammar: An Introduction*. Each nonterminal of G' will be of the form A^m , where A is a nonterminal of G and m is an element of the product monoid.)

Problem 4. Let $\sim$ be the equivalence relation such that $\sim$ -substitutability coincides with k, l -substitutability.

1. Show that $\sim$ is of finite index and each equivalence class of $\sim$ is regular.
2. Assume that a CFG G is in the normal form given by Problem 3. Consider the definition of the characteristic set $CS(G)$ for G of Clark and Eyraud (see the paragraph following Lemma 4 of Yoshinaka). Compare this definition with the definition of K_G preceding Lemma 3 of Yoshinaka.
3. Try to prove Yoshinaka's Lemma 4 using $CS(G)$ instead of K_G . (Cf. Clark and Eyraud's Lemma 14.)

Problem 5. Let $\sim \subseteq \Sigma^* \times \Sigma^*$ be an equivalence relation of finite index such that each of its equivalence classes is regular. Try to establish a learnability result for $\sim$ -substitutable context-free languages.